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CCF501 Cloud Computing Fundamentals

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Assessment – 3

Cloud Application Deployment

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Cloud Application Deployment

Introduction

Nowadays, cloud computing solutions have been increased because of the availability of applications and services. Cloud computing also enhances a better business solution where the data and services can be accessed from anywhere. The cloud service provider is managed by cloud technology where the data can be hosted with monthly or pay-as-you-go. Since

there are several cloud service providers, Amazon Web Service is the most efficient and reliable (Talha et al., 2020, p. 131-136). The AWS service will facilitate efficient resource pooling and security. Similarly, the AWS also permits faster deployment of applications to the user. The purpose of the assessment is to deploy the web application using the AWS cloud service provider.

Background information

Cloud computing has provided an immersive delivery of computing services like storage, databases, analytics, and so on. This allows the user to easily access and manage the resources without any physical infrastructure. Unlike with the traditional IT infrastructure, the organization requires a hardware and software license to use the application. However, cloud computing operates with the pay-as-you-go model where the business can only pay for the resources utilized. The advantage over the traditional IT infrastructure is that cloud computing is cost-effective and easily scalable (Atul Vijay Vishwakarma & Parth Shivraj Dalvi, 2024, p. 294-302). In addition, the cloud provider can also update the system regularly along with the security patches. This minimized the service downtime and the work of the IT staff.

Cloud computing is typically categorized into different service models like IaaS, PaaS, and SaaS. The Infrastructure as a Service provides a computing resource like an operating system, storage, and so on. Whereas the platform as a service will able to manage applications without any infrastructure. The software as a service will deliver an application on a subscription basis (Patel, 2020, p. 85-133). The deployment model of the cloud service that is used in this project is the public cloud. The public cloud deployment model shares the service publicly where all internet users can easily access it. Likewise, AWS (Amazon Web Service) is also a public cloud service provider so that the application can be effectively deployed.

Selected service provider

About Amazon Web Services

Amazon Web Service was selected as a service provider because it has a wide range of solutions for several business needs. The Amazon Web Service has offered several IT services to the public through web services. The services are based on the computing power and data storage. Amazon has also leveraged several solutions according to the business needs without any integration of vendors (Challa et al., 2022, p. 1-2). The AWS also provides scalability and

flexibility to the business organization to scale their performance according to the demands. The infrastructure of the AWS can accommodate global changes without any service disruptions. Similarly, the application and services are available worldwide which indicates the user can able to access the services from any location. The most important feature of selecting this service provider is security which offers several advanced features like access control, data encryption, and compliance standards according to industry operations. The high level of security helps in securing the data thus fostering trust and reliability. In addition, AWS operates in cost-effectiveness where the organization can pay only for the service they use. This model eliminates the large investment and improves the IT infrastructure to deploy several applications effectively (Dubey & Raja, 2023, p. 16-32).

Deployment model

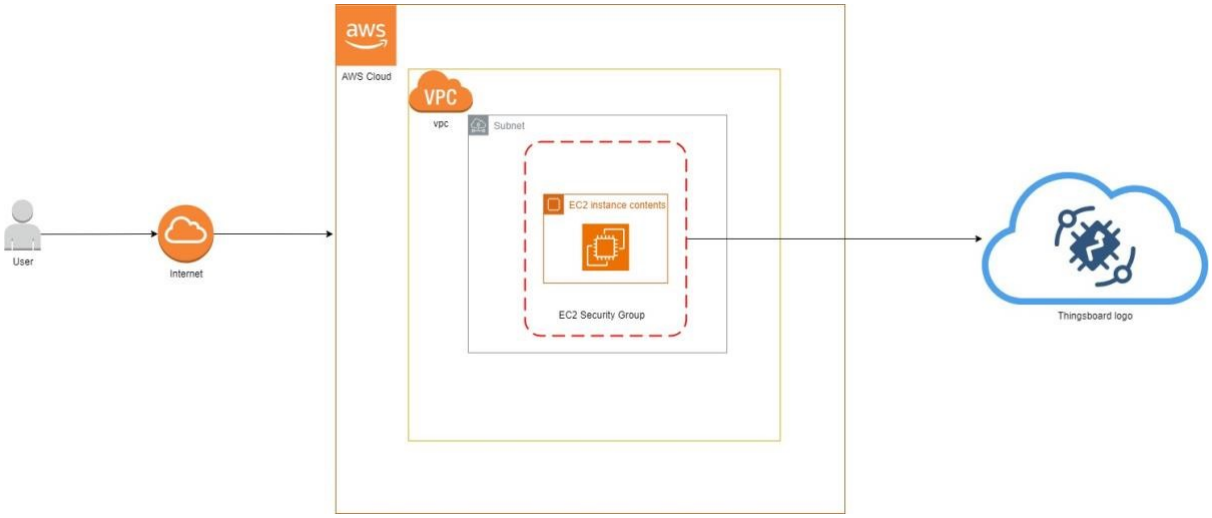
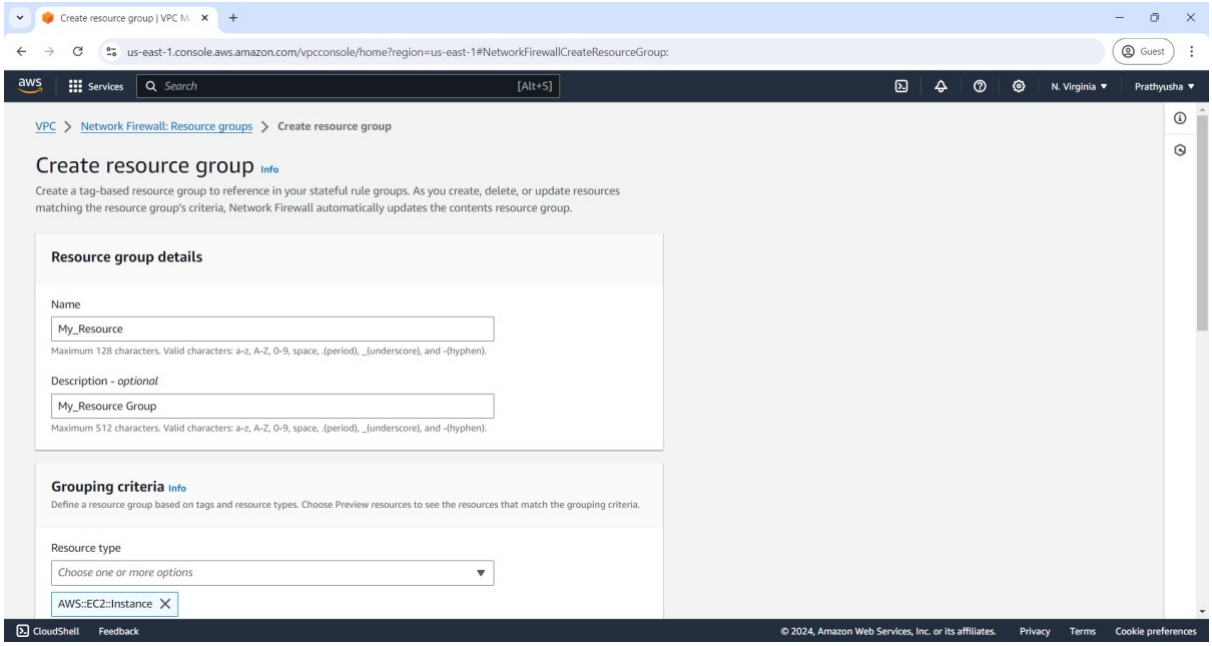


Figure 1: Deployment Model of Web Application in AWS (Self)

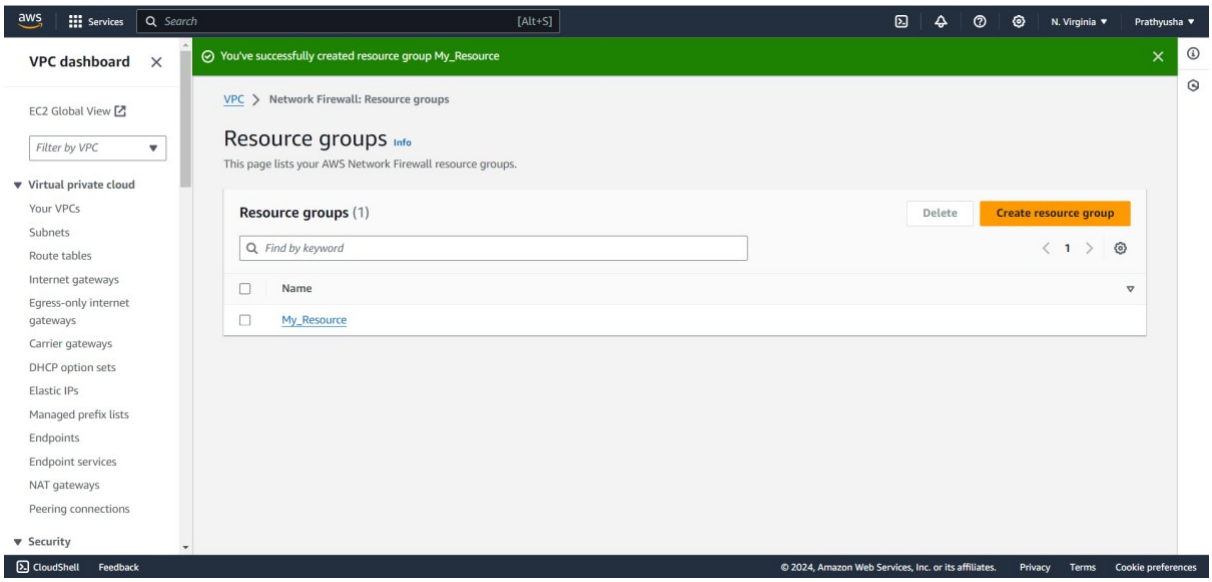
Procedure for Cloud Application Deployment

Creating Resource Group

Initially, the signup procedure is carried out in the Amazon web service to deploy the application. The resource group is created by following AWS Network Firewall Resource group.



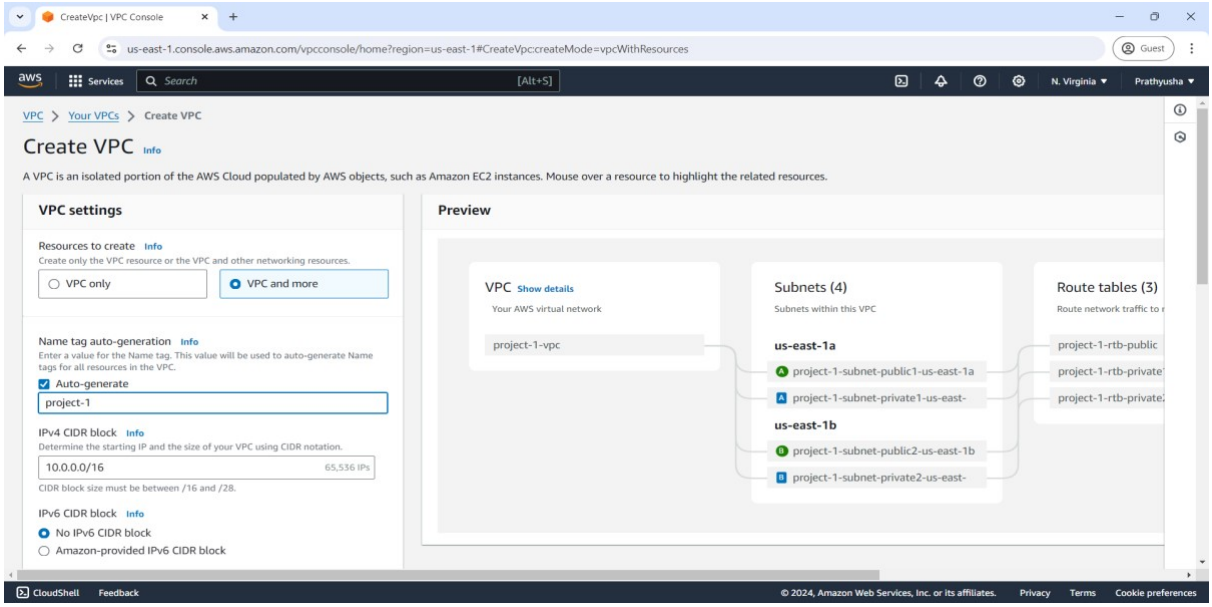
The above screen shows the creation of a resource group where the resource name and description were named. The grouping category is chosen as “AWS:EC2:Instance”.



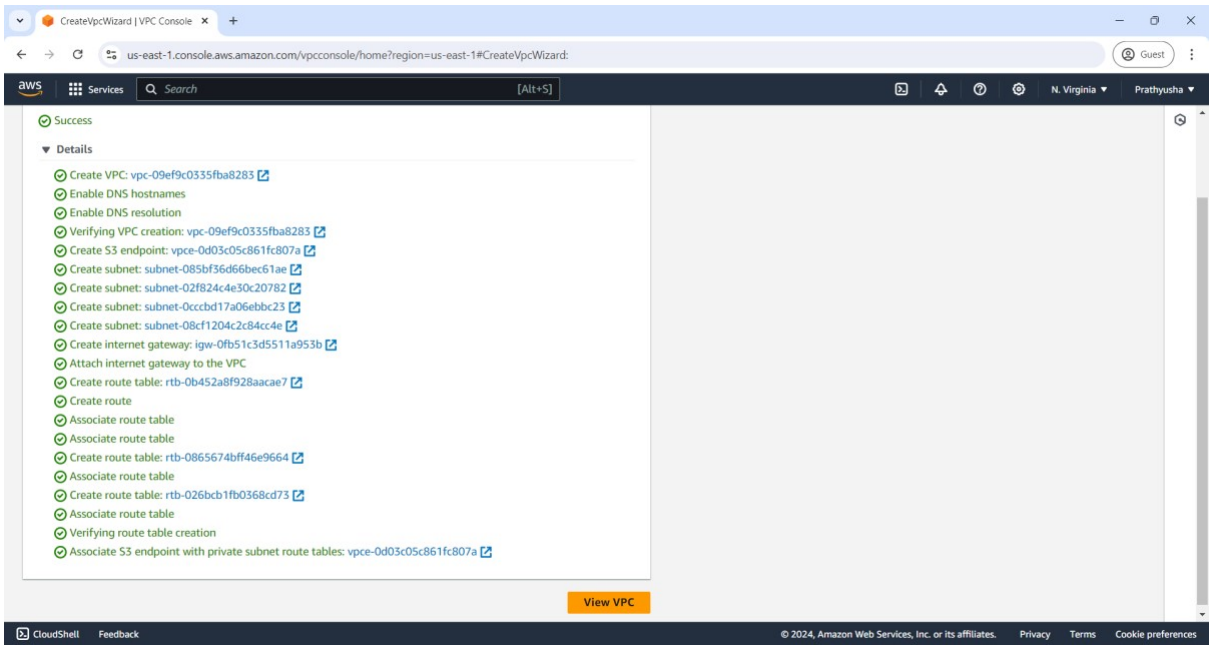
Similarly, the tag name is also provided, and by clicking on the create resource group, the resource group under Network Firewall is created successfully.

Add a virtual network in group

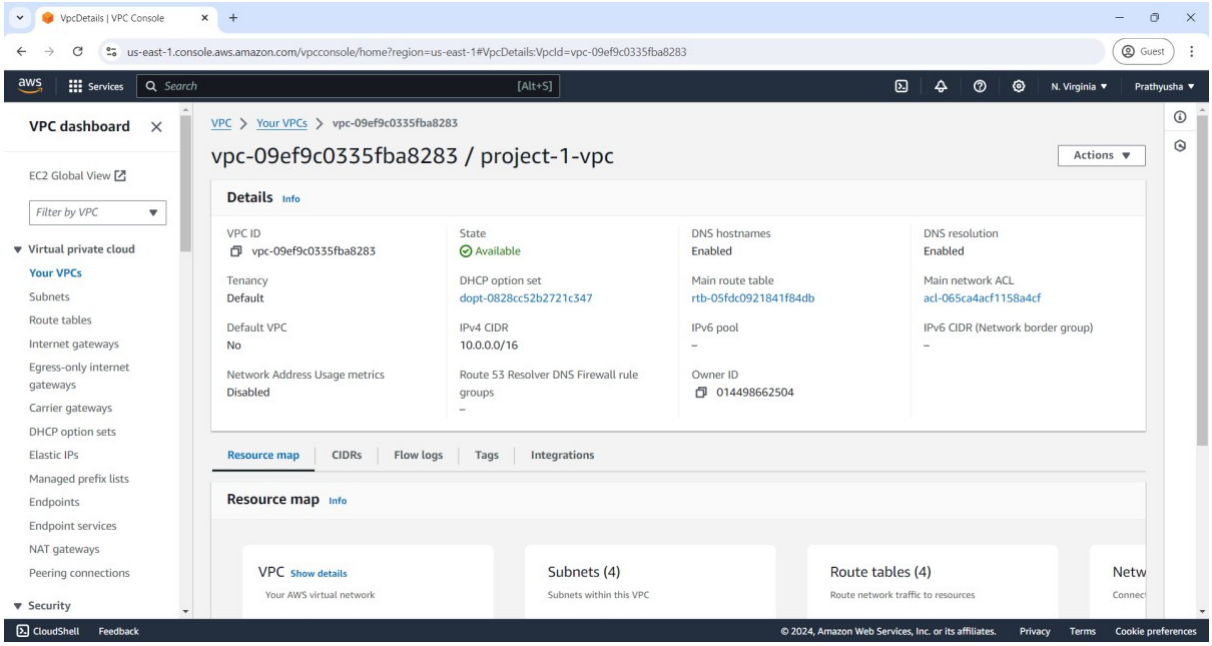
The purpose of the VPC (Virtual Private Network) is to create a private cloud within the public cloud. By selecting the VPC in the service tab, the VPC can be created.



The above screen shows the creation of VPC where the resource to create is selected as “VPC and more”, name as “project-1” and assigned IPv4 address as 10.0.0.0/16. The preview of the VPC shows the connection towards the subnet and default router.



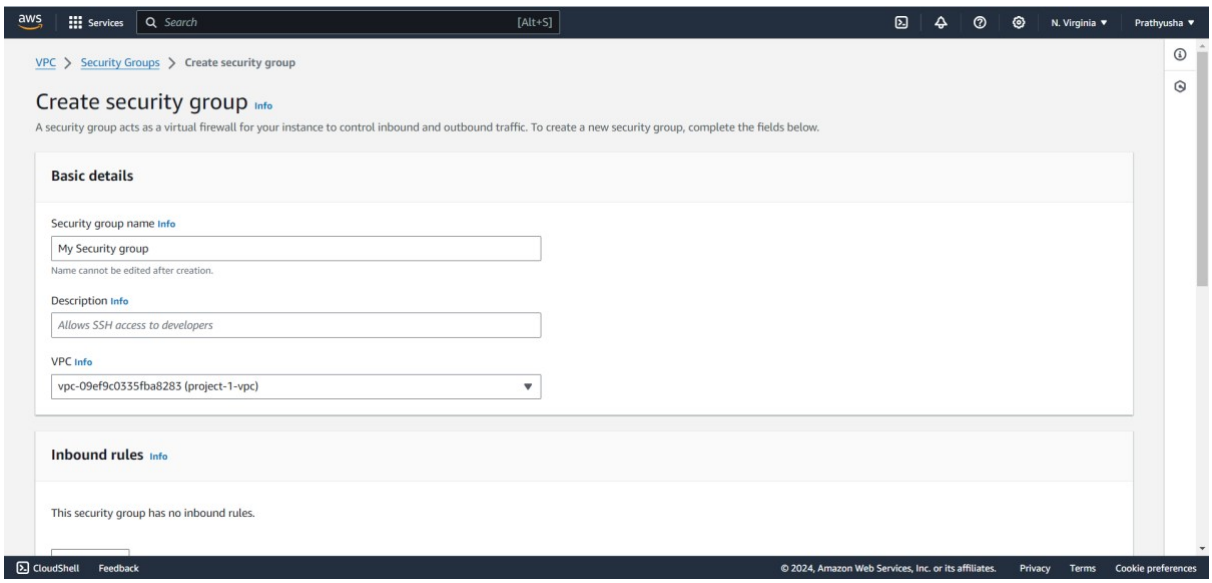
By clicking on “Create VPC”, the VPC is created with the number of subnets, router table, and the S3 endpoints (Dantas et al., 2022).



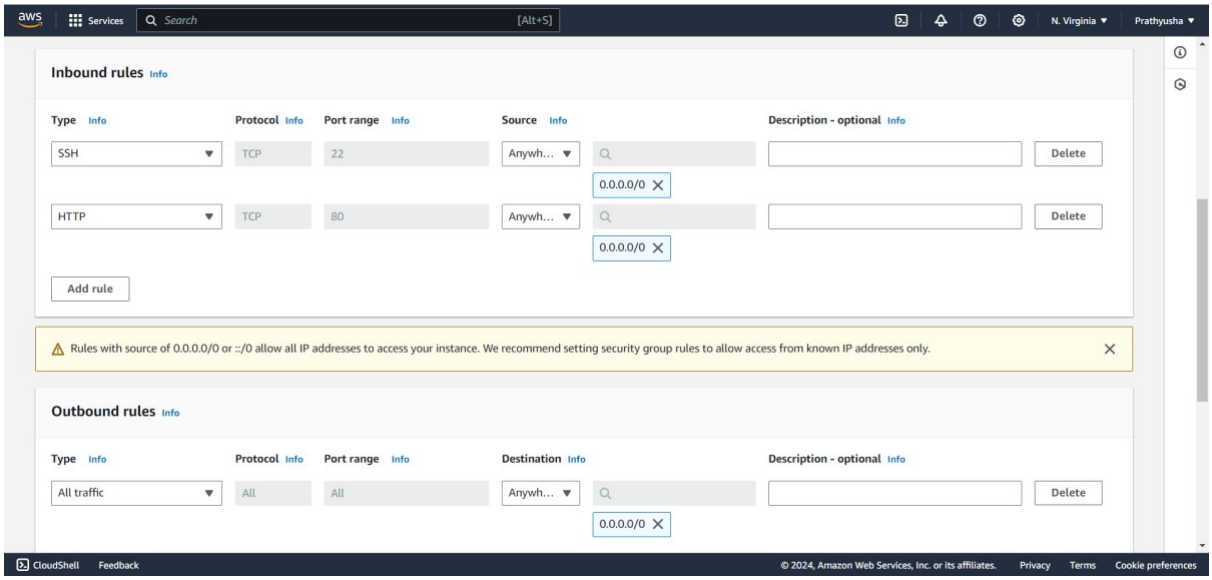
The overview of the created VPC is shown with the different features.

Security Group

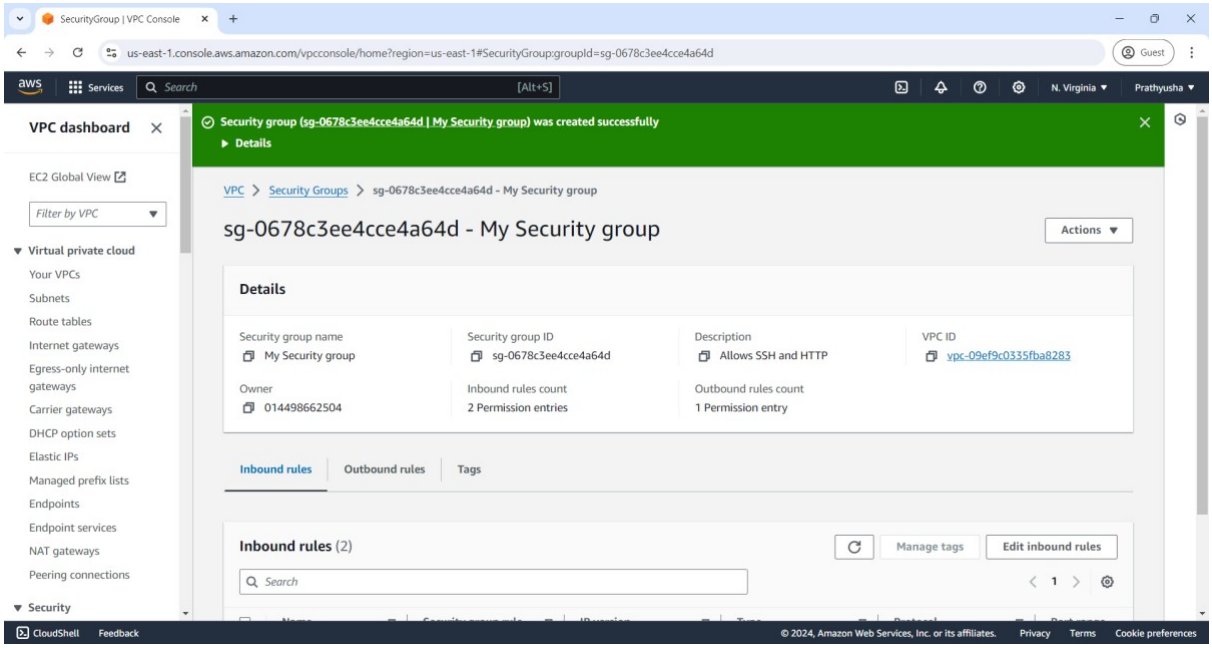
Selecting the security group in the VPC module will able to generate the security group for the application to deploy.



The security group is created with the name “My Security Group” in the description. The VPC is selected as an existing VPC name.



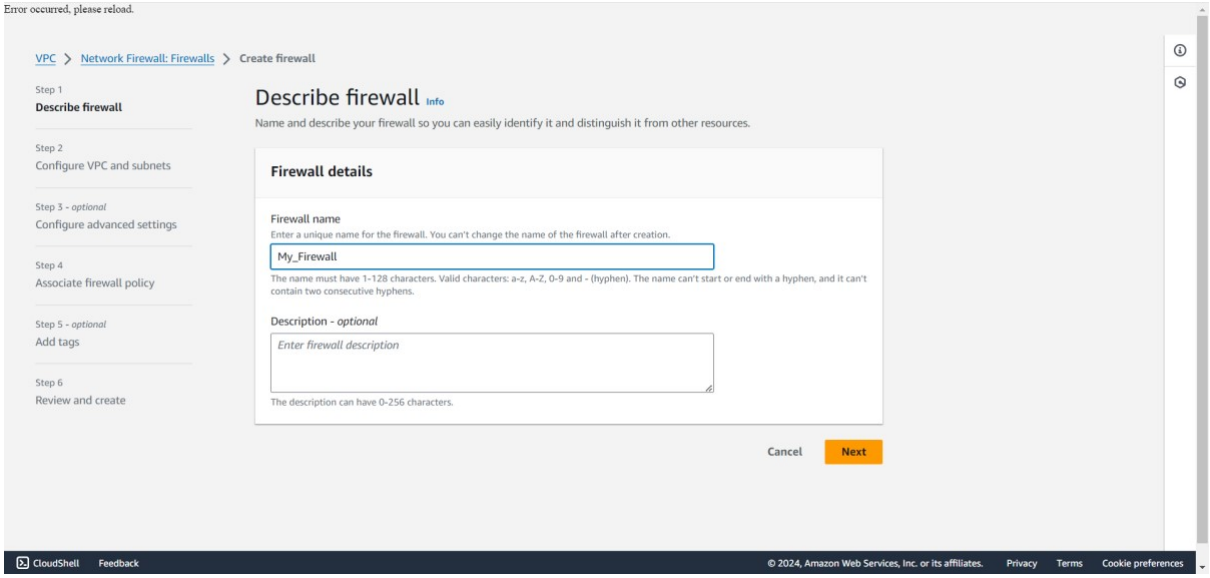
For the selected security group, the inbound and outbound rules were set which is shown above. The inbound rules examined as SSH and HTTP where the source of connection can be established from anywhere.



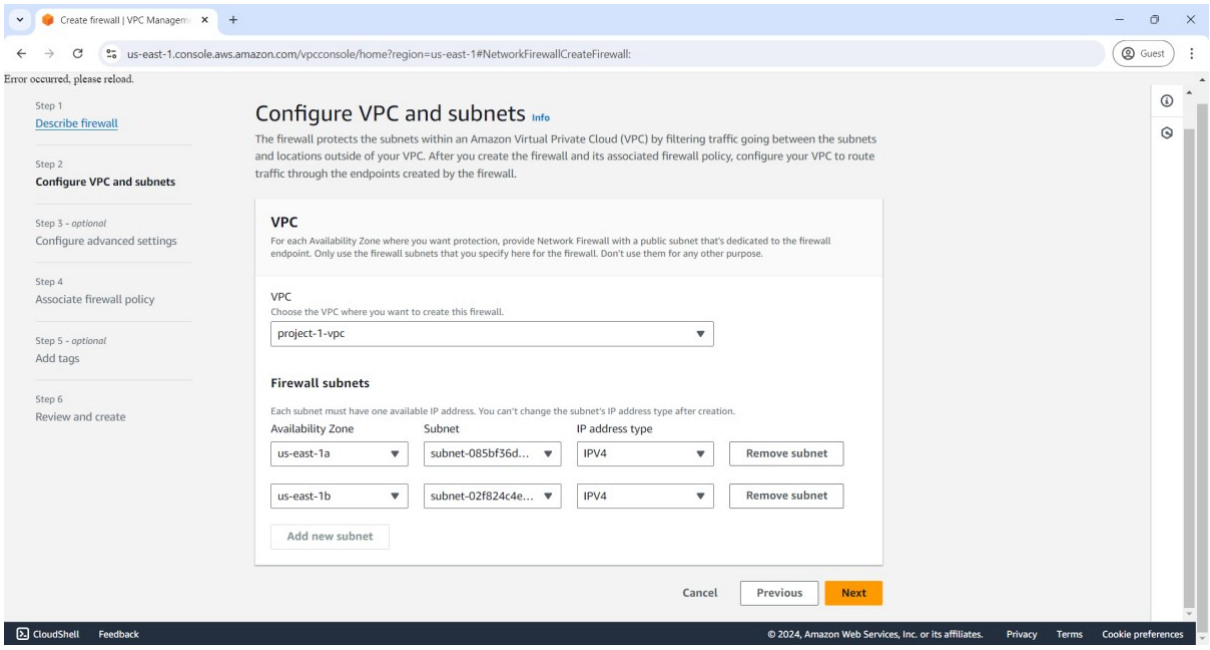
The security group details is shown in the above screen where the it displays the count of inbound and outbound rules with the owner name.

Protect the cloud network with a firewall security policy

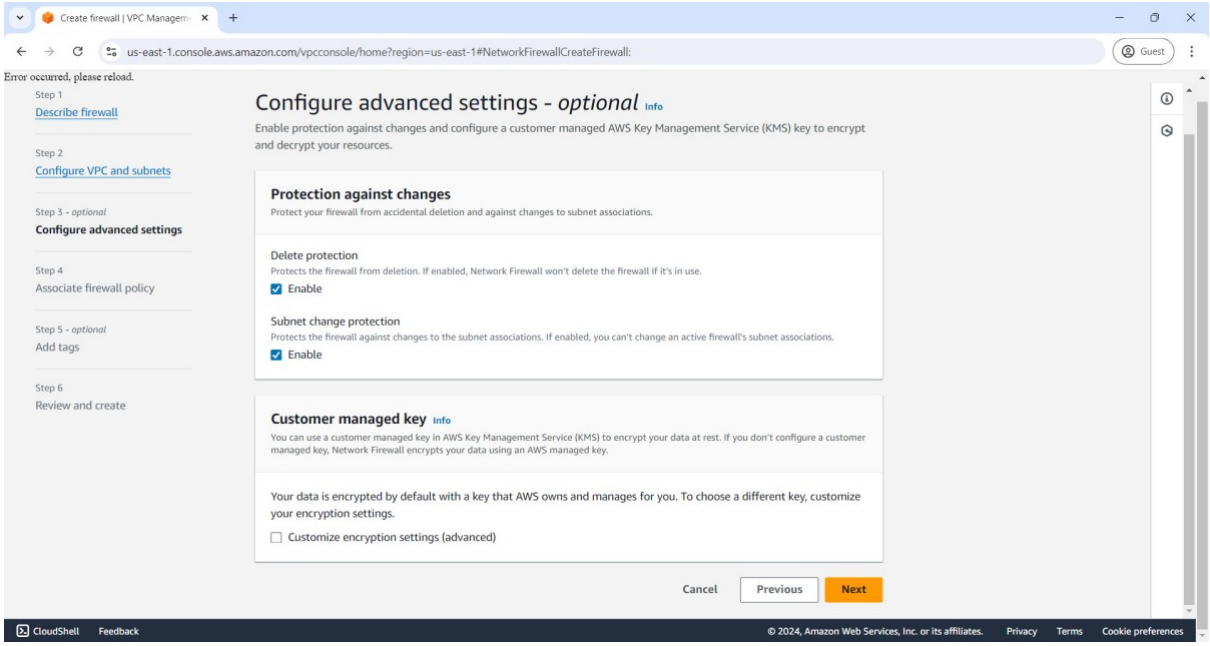
Configuration of a firewall is necessary to secure the connection when the user is connected to the web application. This will protect the entire network within the subnets and Virtual Private Network.



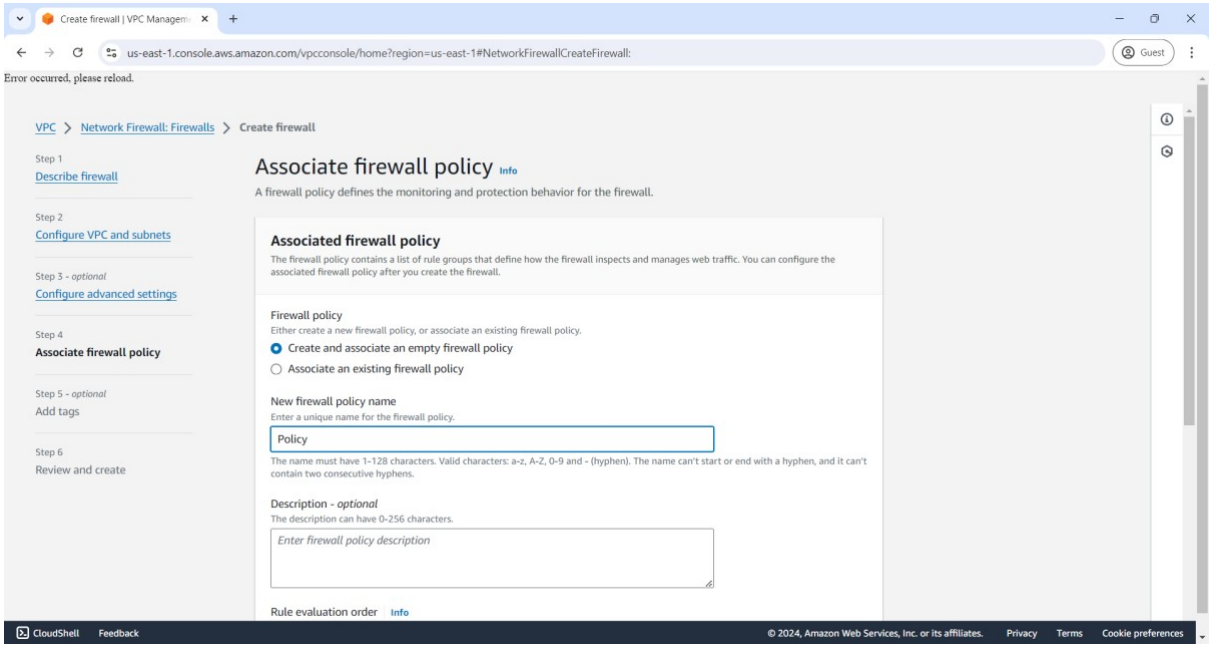
The firewall name is defined as “My_Firewall” which outlines the unique name of the firewall.



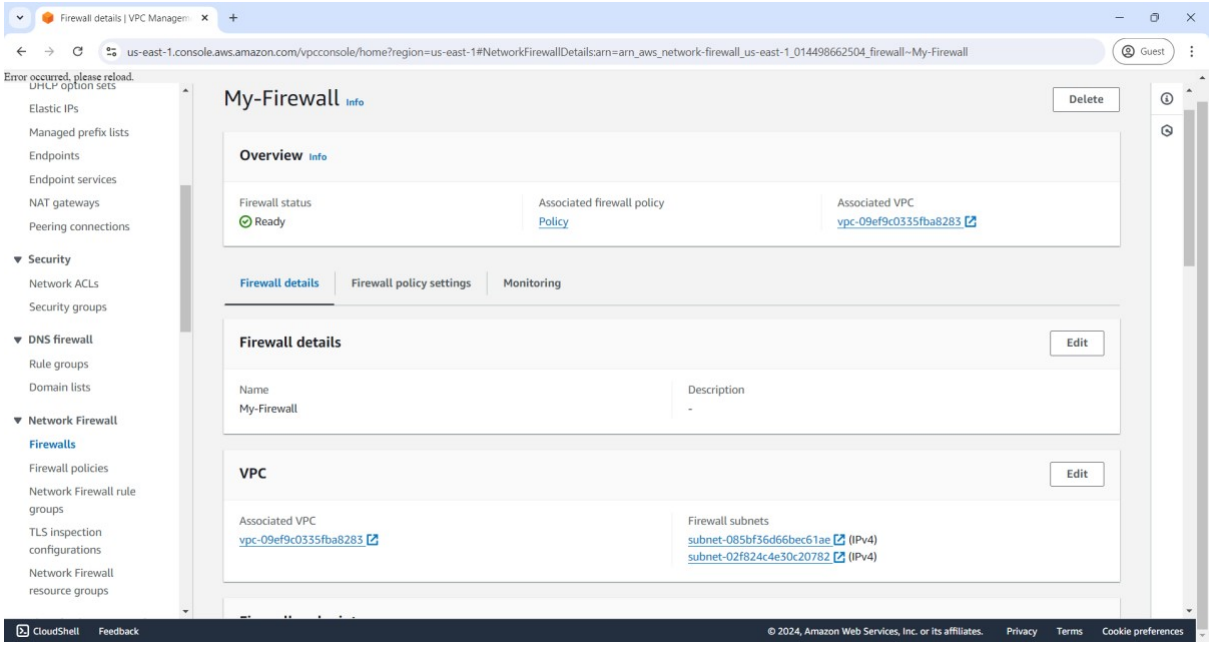
The configuration of VPC and Subnet were made by selecting the appropriate VPC and firewall subnets.



The above screen shows the advanced setting of the firewall configuration where the delete protection and subnet change protection features were enabled (Beuran et al., 2022).



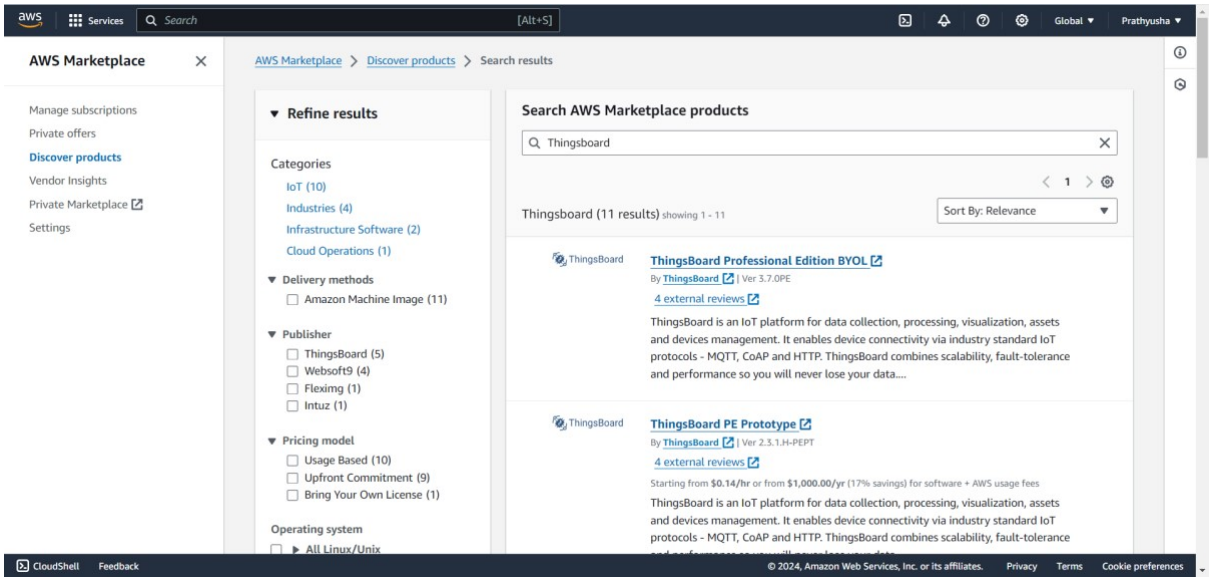
Finally, the associated firewall policy is created by proving the new firewall policy name to manage the web traffic.



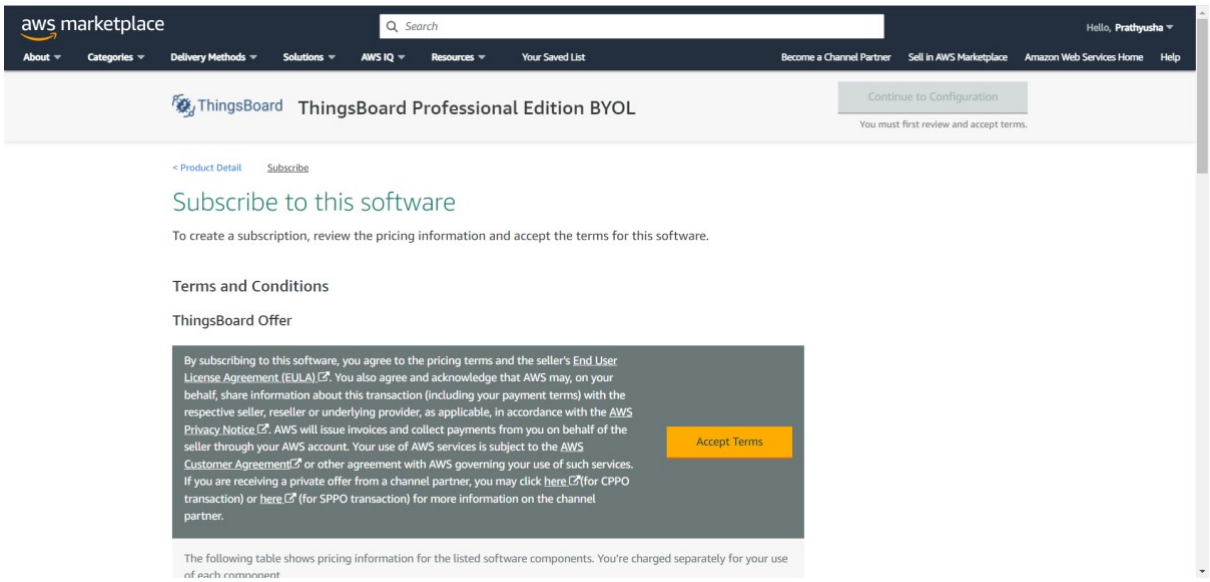
Now the overview of the firewall is shown in the above image. The status of the firewall is ready to use which indicates the web traffic can be monitored effectively with the associated VPC and the firewall subnets.

Deploy of Web application

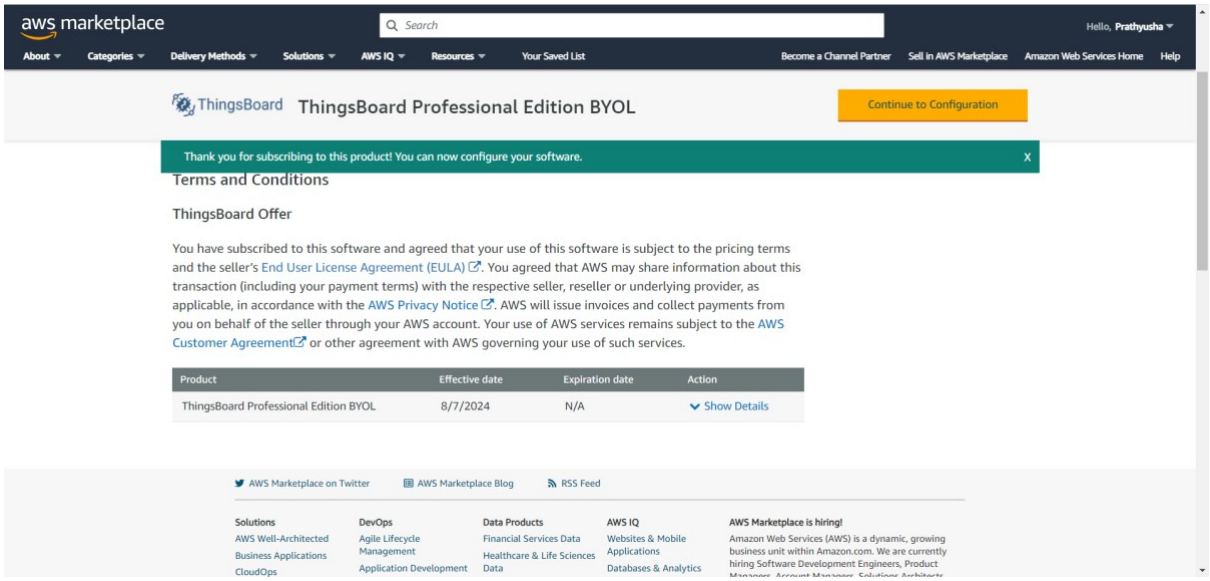
The selected web application to deploy in the AWS cloud service provider is the “ThingsBoard Application”. This application is an open-source solution where the IoT devices can be connected to collect and process the data effectively.



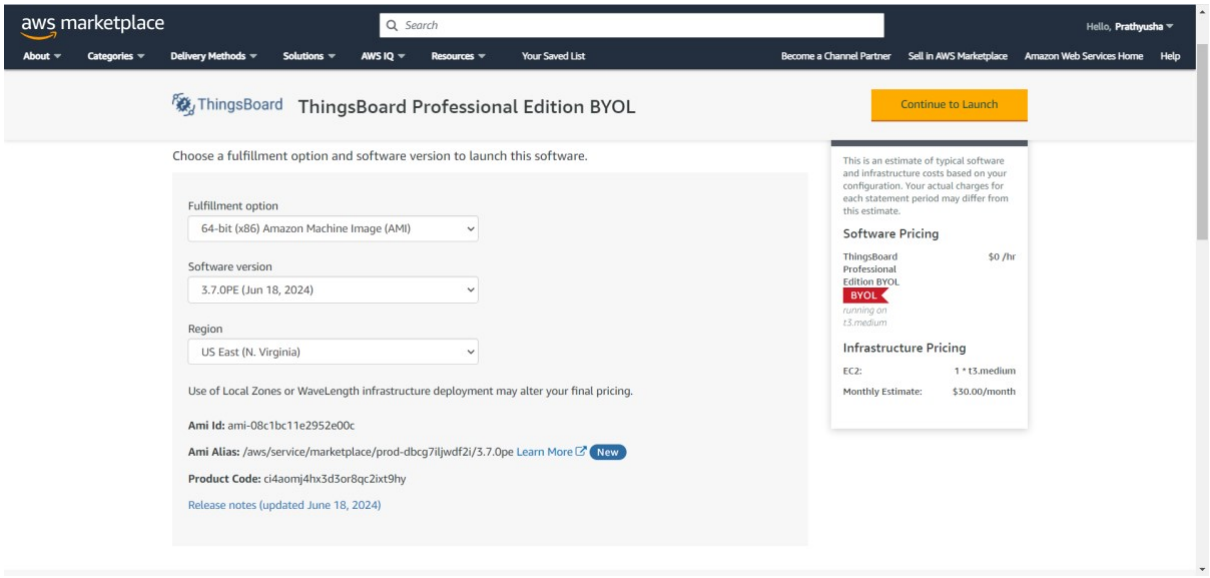
From the AWS Marketplace, ThingsBoard product is discovered effectively with different platforms. Now the “ThingsBoard Professional Edition BYOL” product is selected because it is free of cost to deploy in the AWS cloud server.



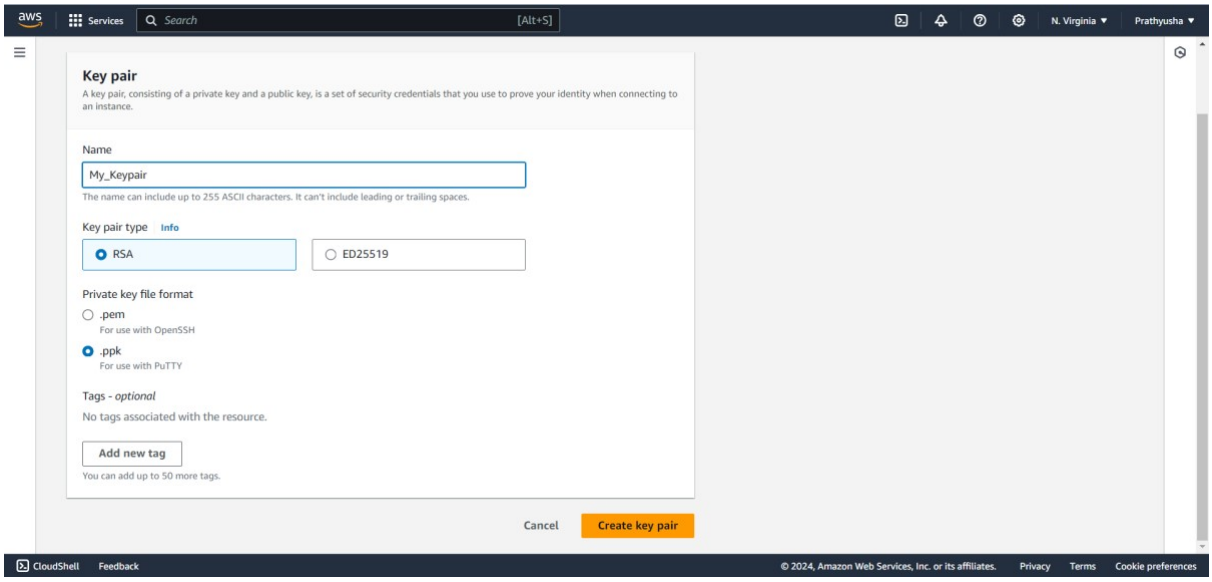
To subscribe to the ThingsBoard software, the acceptance of terms and policy is required which is highlighted in above screen.



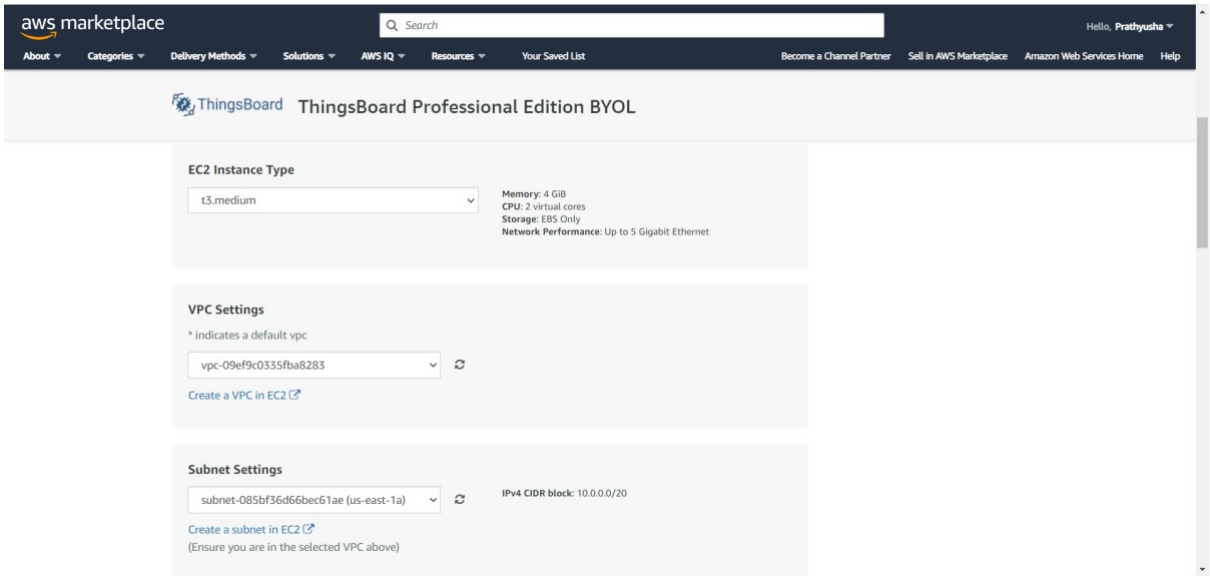
After accepting the terms, the ThingsBoard product creation date was displayed effectively.



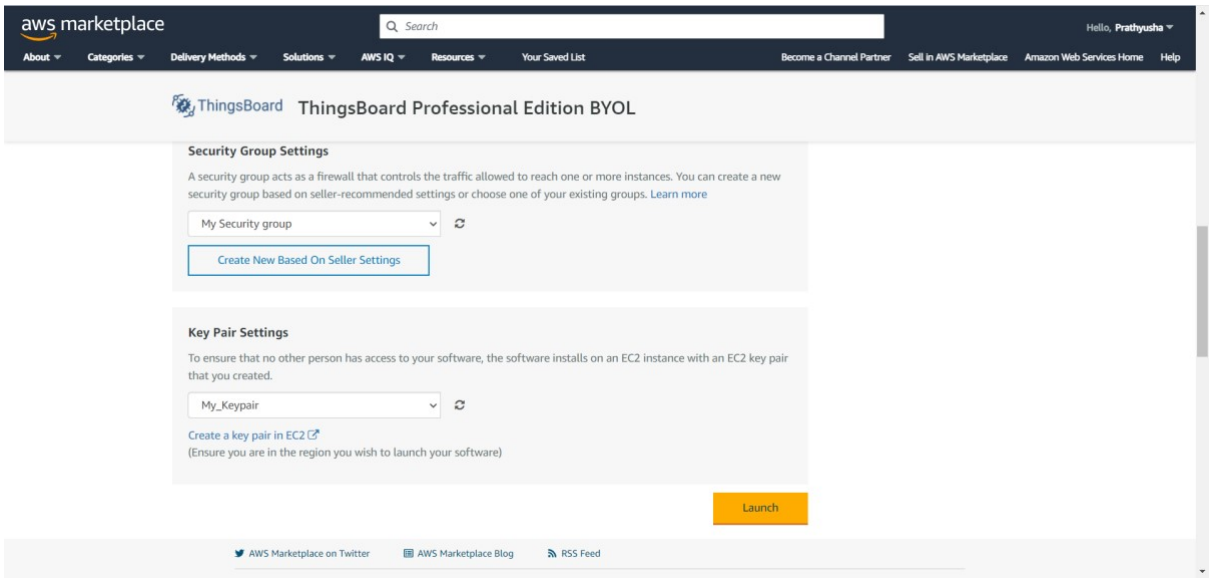
The overall features of the ThingsBoard Professional Edition BYOL application are displayed where the software version and the region of installation were selected (Anand et al., 2023).



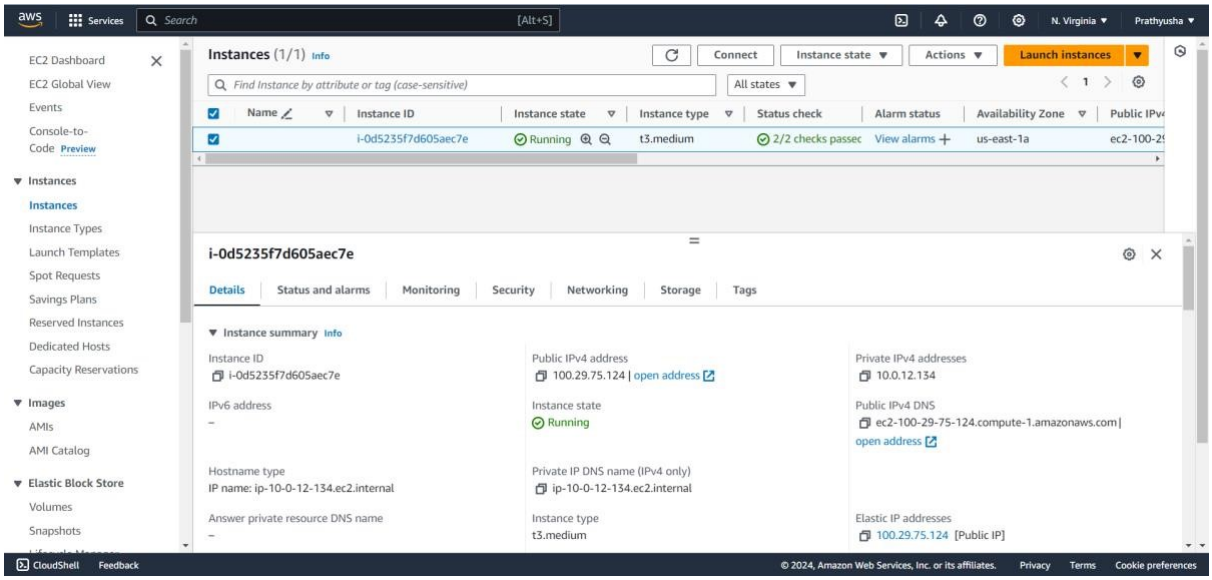
To configure the application, the key pair is required where the key pair is selected as RSA, and the file format as “.ppk”.



The EC2 instance type to deploy the application is selected as “t3.medium” which contains 2 virtual cores and a network performance of 5 GB Ethernet. The VPC and subnet setting is selected with the created VPC.

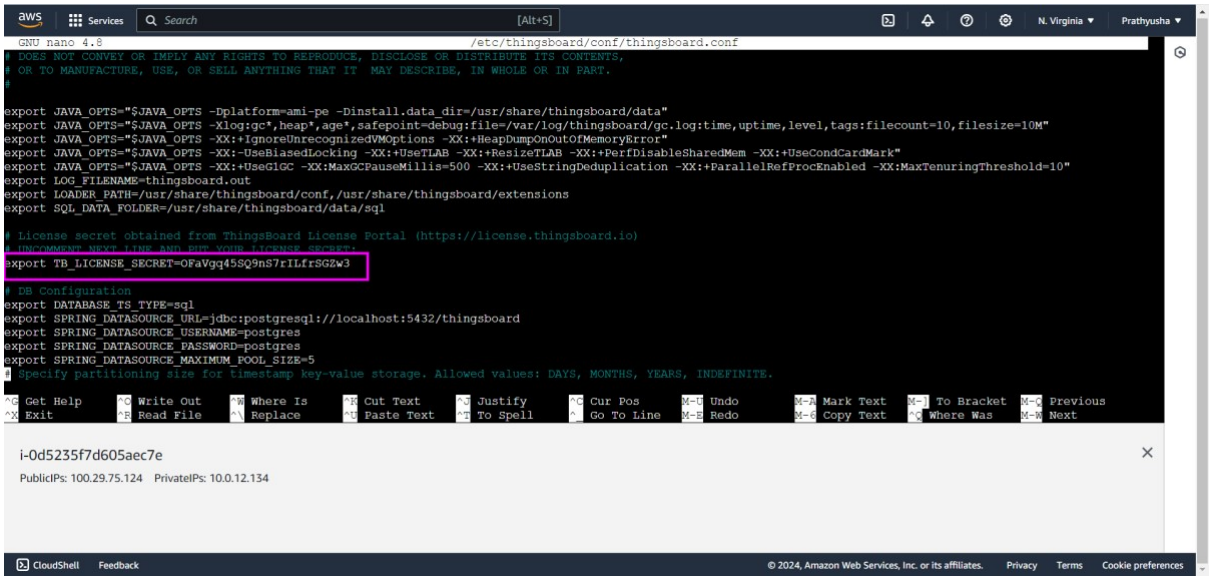


Similarly, the security group setting is featured in order to control all the network traffic for the instance. To ensure the secure access to the application, the keypair is also selected.



After the successful launch of the application, the instance is created with an instance ID, type, public IPv4 address, private address and so on (Bagai, 2024, p. 102-110).

The above command explains the configuration of the Thingsboard application.



The license key for the Thingboard application is entered in the configuration file.

```
root@ip-10-0-12-134:~# sudo nano /etc/thingsboard/conf/thingsboard.conf
root@ip-10-0-12-134:~#
root@ip-10-0-12-134:~#
root@ip-10-0-12-134:~#
root@ip-10-0-12-134:~#
root@ip-10-0-12-134:~#
root@ip-10-0-12-134:~#
root@ip-10-0-12-134:~# sudo service thingsboard start
root@ip-10-0-12-134:~#
```

i-0d5235f7d605aec7e

PublicIPs: 100.29.75.124 PrivateIPs: 10.0.12.134

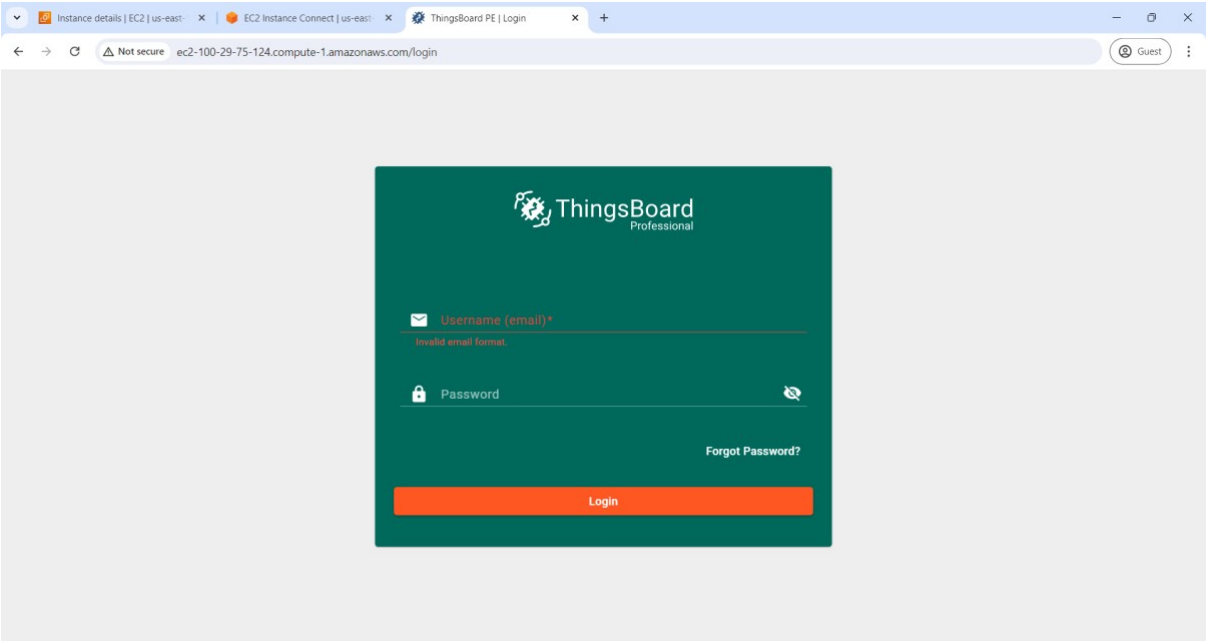
Now, the Thingsboard application is started using the above command.

```
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

root@ip-10-0-12-134:~#
root@ip-10-0-12-134:~#
root@ip-10-0-12-134:~#
root@ip-10-0-12-134:~#
root@ip-10-0-12-134:~#
root@ip-10-0-12-134:~#
root@ip-10-0-12-134:~# sudo nano /etc/thingsboard/conf/thingsboard.conf
root@ip-10-0-12-134:~# sudo nano /etc/thingsboard/conf/thingsboard.conf
root@ip-10-0-12-134:~#
root@ip-10-0-12-134:~#
root@ip-10-0-12-134:~# sudo /usr/share/thingsboard/bin/install/install.sh --loadDemo
OpenJDK 64-Bit Server VM warning: Option UseBiasedLocking was deprecated in version 15.0 and will likely be removed in a future release.

 :: ThingsBoard Professional Edition ::       (v3.7.0PE)
```

i-0d5235f7d605aec7e

PublicIPs: 100.29.75.124 PrivateIPs: 10.0.12.134

The database of the ThingsBoard application is initialized effectively.

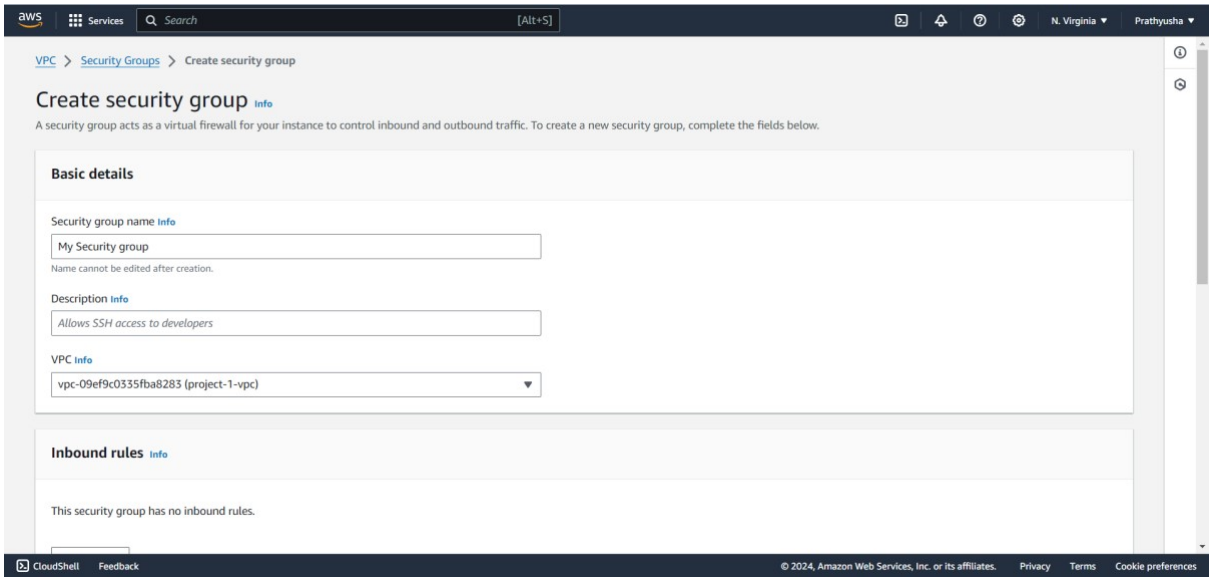


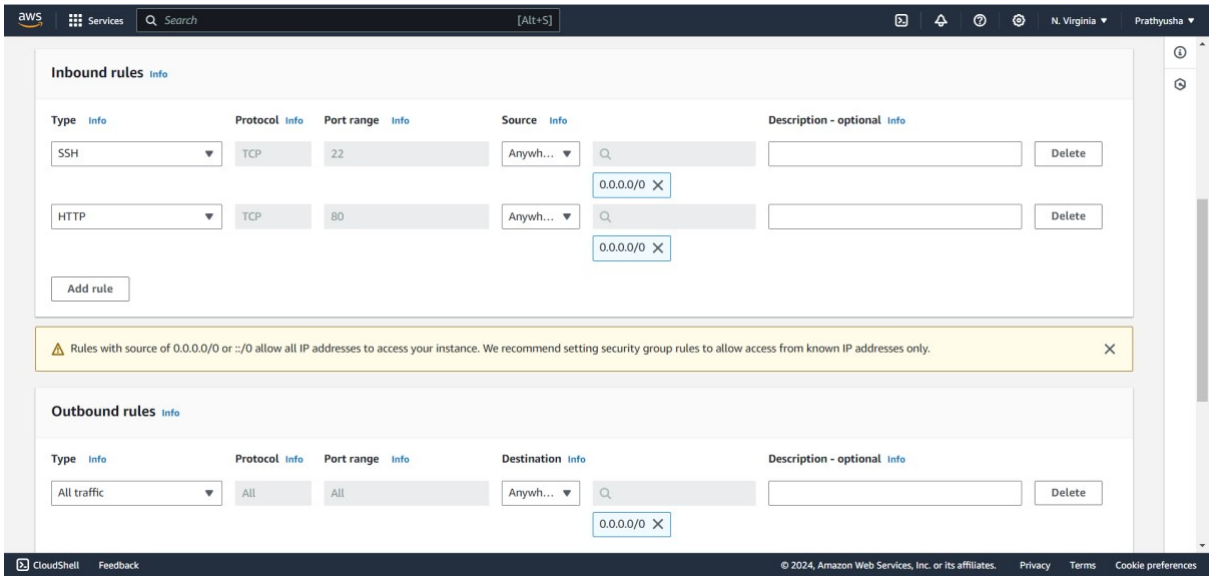
By connecting to the browser, the ThingsBoard application is deployed using the IPv4 public DNS address which is running in the EC2 instance.

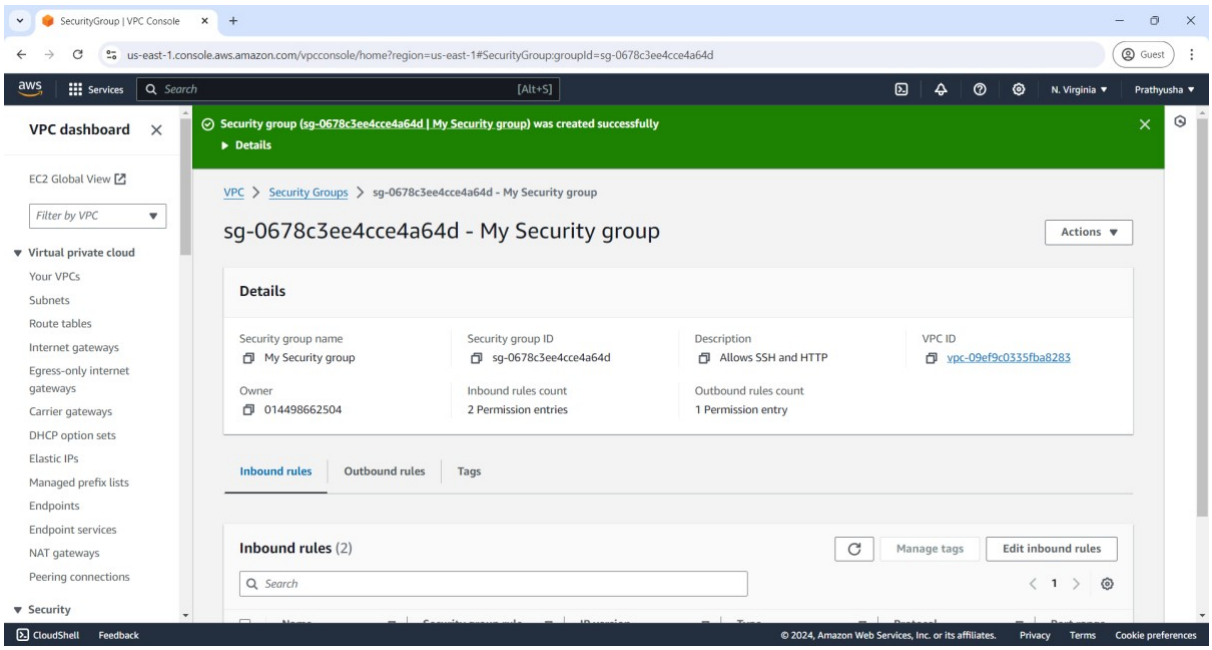
Link: http://ec2-100-29-75-124.compute-1.amazonaws.com/login

Cloud security policies

A cloud security policy has been developed to protect the cloud network so that unauthorized access can be effectively monitored. The below screens are the list of security policies for the deployment of the application.







Analysis of an application

With the help of cloud computing technology, the performance of the web application can be enhanced easily. There are several challenging factors like large volume storage which impacts the business performance and operations. In addition, the data which is collected from the application must be managed securely within the AWS infrastructure (T & Kumar, 2021). Cloud security is also an important factor where the AWS achieves in establishing proper integrity and availability. Because of the open source community, the application requires a proper security mechanism and frequent updates.

Conclusion

The enhancement of cloud computing helps the business environment to quickly process the operation through web application deployment. The report provides a brief background introduction about cloud computing, deployment model, and service providers. According to the analysis, the AWS (Amazon Web service) cloud technology is selected in order to deploy the web application. This cloud technology provides significant benefits to security policies and data storage. Due to cost-effectiveness, the cloud user can get high-speed services effectively. The significance of the AWS is network security where the firewall protection is enhanced to secure the connection. The project also includes a virtual network where the VPC facilitates a secure connection to the application. In addition, the cloud firewall policy is deployed in the virtual network to secure the connection. The deployment of the ThingsBoard web application is enhanced because it is open-source and reliable. The deployment technique provides flexibility to access the application when the user gets permits. The extensive feature of the AWS is about the efficiency of the system where the development process can be moved forward effectively. Therefore, AWS provides an efficient service to deploy the web application in the cloud along with security policies and firewalls.

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